

dr. Tom Lauwaerts

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Dutch Native · English High proficiency · French Elementary · Chinese Novice

Work experience

Post-doctoral Researcher

Vrije Universiteit Brussel · 2025 - present

PhD Researcher: Programming Languages

Ghent University · 2021 - 2025 (4 years)

Research into debugging and testing tools with a strong formal foundation.

Teaching Assistant

Ghent University · 2021 - 2025 (4 years)

Courses: *Fundamenten van programmeertalen*, *Logisch programmeren*.

Embedded Software Developer (Internship)

Creative Therapy (startup) · July - August 2020

Education

PhD in Computer Science · 2021 - 2025

Ghent University, TOPL Lab

BSc. and MSc. in Computer Science · 2016 - 2021

Ghent University, Faculty of Sciences

Highlighted research output



MIO: Multiverse Debugging in the Face of Input/Output

2025-10 | [Tom Lauwaerts](#), [Maarten Steevens](#), [Christophe Scholliers](#) | *Proceedings of the ACM on Programming Languages (PACMPL)*, Volume 9, Issue OOPSLA2

I presented this work at the main track of the OOPSLA 2025 conference.

Academic services

IC00OLPS workshop co-organizer · 2026 - 2027

Co-organizing the IC00OLPS'26 and IC00OLPS'27 workshops.

ECOOP student volunteer chair and local organizer · 2026

Part of the VUB team organizing ECOOP'26 in Brussels.

DEBT workshop co-organizer · 2025

Co-organizing the DEBT'25 workshop.

Programming artifact evaluation committee member · 2024 - 2027

AEC member for the Programming conference and journal (vol. 9, 10, 11, 12).

Volunteer work

Science communication and outreach · 2021 - present

Alumniday UGent, Dag van de Wetenschap, Comp Sci Olympiad ...

Open-source software contributor · 2021 - present

 [WARDuino](#) ·  [latch](#) ·  [personal projects](#)

Honors

ISSTA'23 Research competition: 3rd place · 2023

International ACM Research competition at ECOOP/ISSTA 2023 3rd place medal.

Concolic Multiverse Debugging

2026-06 | Maarten Steevens, **Tom Lauwaerts**, Christophe Scholliers | *Special Issue of the 40th European Conference on Object-Oriented Programming (ECOOP 2026)*

An early version of this work was presented by myself at the final of the ISSTA'23 Research competition.

MIO: Multiverse Debugging in the Face of Input/Output

2025-10 | **Tom Lauwaerts**, Maarten Steevens, Christophe Scholliers | *Proceedings of the ACM on Programming Languages (PACMPL), Volume 9, Issue OOPSLA2*

I presented this work at the main track of the OOPSLA 2025 conference.

Foundations for Constrained Debugging Techniques

2025-08 | **Tom Lauwaerts** | *PhD dissertation (Universiteit Gent)*

Latch: Enabling large-scale automated testing on constrained systems

2024-12 | **Tom Lauwaerts**, Stefan Marr, Christophe Scholliers | *Science of Computer Programming Journal*

I presented this work during a poster session at the ISSTA'23 conference.

WARDuino: An embedded WebAssembly virtual machine

2024-06 | **Tom Lauwaerts**, Robbert Gurdeep Singh, Christophe Scholliers | *Journal of Computer Languages*

Presented at the International Conference on the Art, Science, and Engineering of Programming 2022, during a poster session and a dedicated demonstration.

Out-of-Place Debugging on Constraint Devices with the EDWARD Debugger (Demo)

2023-07 | **Tom Lauwaerts**, Carlos Rojas Castillo, Elisa Gonzalez Boix, Christophe Scholliers | *Short paper (DEBT'23)*

Demo presentation at the first Workshop on Future Debugging Techniques (DEBT'23).

Demo: Debugging Constraint Devices with EDWARD

2023-06 | **Tom Lauwaerts**, Carlos Rojas Castillo, Elisa Gonzalez Boix, Christophe Scholliers | *Short paper (MobiSys'23)*

Demo and poster presentation at the 21st ACM International Conference on Mobile Systems, Applications, and Services (MobiSys'23).

Event-Based Out-of-Place Debugging

2022-09 | **Tom Lauwaerts**, Carlos Rojas Castillo, Robbert Gurdeep Singh, Matteo Marra, Christophe Scholliers, Elisa Gonzalez Boix | *Conference paper (MPLR'22)*

I presented this work at the main track of the 19th International Conference on Managed Programming Languages & Runtimes (MPLR'22).